

Open Elective :

Cyber & Digital Safety

Course Objectives (CO)

- CO 1.** To introduce the foundational concepts of cyber safety and digital privacy.
- CO 2.** To create awareness of common online threats and best practices for digital hygiene.
- CO 3.** To equip students with skills for safe usage of social media, devices, and the internet.
- CO 4.** To develop a mindset for ethical digital behavior and responsible technology use.
- CO 5.** To explore emerging cyber risks including AI, deepfakes, and the role of legislation.

Course Outcomes (OC)

After successful completion of this course, students would be able to

OC 1. Understand key concepts in cyber security and differentiate between threats and risks.

OC 2. Apply safe browsing practices and protect their digital identity across platforms.

OC 3. Identify and avoid common cyber scams, phishing attacks, and fake news.

OC 4. Safeguard personal devices, use communication tools, and manage passwords. e communication tools, and manage passwords.

OC 5. Understand the ethical concerns and emerging trends in AI-driven cyber threats.

1 Vertical: Open Elective

2 Type: Theory

3 Credits: 2 credits

4 Marks Allotted 50 Marks

5 Internal Continuous Assessment: 40% (20 marks)

Continuous Evaluation through Class Tests and Assignments

Test on Module 1: 10 marks

Test on Module 2: 10 marks

Average of 2 Class Tests: 10 marks

Assignment on Module 1: 5 marks

Assignment on Module 2: 5 marks

Total of 2 Assignments: 10 marks

Total: 10+10 = 20 marks

6 Semester End Examination :60% (30 marks)

Evaluation through: A Semester End Theory Examination of 1 hour duration for 30 marks as per the paper pattern given below.

Format of Question Paper:

Total Marks: 30

Duration: 1 Hour

Question	Based On	Options	Marks
Q. 1	Module 1	Any 2 out of 4	10
Q. 2	Module 2	Any 2 out of 4	10
Q. 3	Module 1 & 2	Any 2 out of 4	10

Modules:

Module 1: Fundamentals of Digital Safety & Online Behavior

Introduction to Digital Safety and Cyber Security: What is Digital Safety?, Everyday importance of staying safe online, Cyber Security vs. Digital Safety, Ethical and responsible use of technology

Common Online Threats and How to Avoid Them: Types of online threats: viruses, phishing, scams, Understanding social engineering and digital fraud, Case examples of real-world cyber attacks

Safe Internet and Social Media Use: How to browse safely: HTTPS, safe websites, online shopping tips, Social media privacy settings, Digital identity and managing your online presence, Deepfakes and misinformation, Screen time, mental health, and digital well-being

Online Safety for Children and Teenagers: Common risks for young users, Using parental controls and safe browsing tools, Identifying and addressing cyberbullying, Teaching children responsible internet use

Module 2: Personal Device Security, Privacy & Emerging Trends

Securing Devices and Networks: How to keep your phone and laptop safe, Safe use of public Wi-Fi and VPNs, Protecting smart devices (IoT) at home, Backing up your data securely

Building Good Digital Habits (Cyber Hygiene): Creating strong passwords and using multi-factor authentication, Keeping software updated and avoiding unsafe downloads, Spotting fake news and online hoaxes

Understanding Data Privacy: What is personal data and why it matters, Data protection laws in simple terms (like GDPR), Using privacy-friendly tools (browsers, search engines, messaging apps), What to do if a data breach happens

Cyber Safety at Work and Emerging Technologies: Staying safe while working online or remotely, Secure email and workplace communication, How AI is used in cyber security – the good and the bad, Deepfakes, AI-based scams, and how to stay alert, Introduction to how blockchain can help in digital security

Text Books

- 1. Security in the Cyber Age:** An Introduction to Policy and Technology, John E. Savage, Derek S. Reveron, Cambridge University Press, 2023
- 2. Cyber Security and Personal Data Awareness:** Leverage Personal Data safety in Cyber Threat environment, Selvaraj G, Notion Pres, 2024
- 3. Cyber and Digital Safety:** Fundamentals and Best Practices, Maurya R K, SYBGEN Learning, 2025

Reference Books

- 1. Cyber Safety for Everyone:** A comprehensive guide to online safety, JaagoTeens, BPB Publications, 3rd Edition, 2024
- 2. The Basics of Cyber Safety:** Computer and Mobile Device Safety, John Sammons (Author), Michael Cross MD, Syngress, 2016
- 3. Cybersecurity for Everyone,** David B. Skillicorn, Routledge, CRC Press, 2022
- 4. Cybersecurity for Dummies,** Joseph Steinberg, Wiley, 2020

Example 5: AI-Based Voice Scam (Current Trend)

Scammers use artificial intelligence to clone a person's voice and call family members requesting urgent money.

Digital Safety Practice:

Verify the caller through another phone call or message.

Do not transfer money immediately.

Use a family verification code or question.

Cyber Security Measures:

Telecom providers deploy AI to detect scam calls.

Banks monitor suspicious transactions.

Voice authentication systems help identify legitimate users.

Digital Safety refers to the practices and precautions taken to protect :

Individuals

Devices

Information

while using digital technologies and the internet

Digital Safety involves safeguarding :

Personal data

Avoiding cyber threats

Promoting responsible online behavior

Importance of Digital Safety:

Protects personal information.

Prevents identity theft

Ensures safe online communication.

Maintains privacy on social media.

Promotes responsible online behaviour

****Identity theft or piracy** occurs when someone uses another's personal identifying information, like their

Name

Credit card number

Aadhar card number etc.

, without their permission, to commit fraud or other crimes

What is Cyber?

Cyber refers to anything related or connected to

- Computers,**
- Digital devices,**
- Networks,**
- Internet**

It is commonly used to describe activities, technologies, and security in the online world.

Example: cyber security, cyber attacks, cyber bullying, cyber crime and cyber space

Cyber Security is the protection of:

Computer systems

Networks

Software

Data

from unauthorized access, cyber attacks, damage, or theft

Cyber Security aims to ensure

Confidentiality

Integrity

Availability of information

Cyber security – Protecting computers, networks, and data from hackers and cyber attacks.

Cyber crime – Crimes committed using computers or the internet, such as hacking, phishing, or online fraud.

Cyber bullying – Harassing or threatening someone using digital platforms like social media or messaging apps.

Cyber Attack – An attempt to damage, steal, or gain unauthorized access to computer systems or data.

Cyber Space – The virtual world where people communicate and share information through the internet.

Importance of Cyber Security

Protects systems from cyber attacks.

Prevents financial losses.

Safeguards sensitive information.

Ensures business continuity.

Maintains trust in digital services.

Cyber Safety

Cyber Safety is the practice of using computers ,mobile devices and the internet in a safe and responsible way to protect

yourself

your personal information

your devices

from online threats such as

hacking

viruses

scams

cyber bullying.

Safety measures

- Create **strong passwords** and do not share them with others.
- Keep your **OTP, PIN, and banking details private**.
- Enable **Two-Factor Authentication (2FA)** on important accounts.
- Do not click on **unknown links or suspicious emails**.
- Install and update **antivirus software**.
- Keep your phone, computer, and apps **updated**.
- Use **secure Wi-Fi networks** when accessing sensitive information.
- Take **regular backups** of important files.
- Download apps only from **trusted sources**.
- Log out of accounts when using **shared devices**.

Differences

Digital Safety

- Focuses on the safe and responsible use of digital technologies.
- User-centric.
- Includes online behavior and digital well-being.
- Covers privacy, ethics, and responsible internet use.
- Emphasizes awareness and safe practices.
- Concerned with individuals' daily digital activities.

Cyber Security

- Focuses on protecting digital systems and networks from cyber attacks.
- Technology-centric.
- Includes technical security mechanisms.
- Covers protection of networks, systems, applications, and data.
- Emphasizes prevention, detection, and response to attacks.
- Concerned with securing organizational and personal digital infrastructure.

Example 1: Phishing Email (Digital Safety)

A student receives an email claiming to be from their bank asking them to click a link and verify their account details. The email looks genuine but is fraudulent.

Digital Safety Practice:

- Do not click the suspicious link.
- Check the sender's email address.
- Visit the bank's official website directly instead of using the email link.
- Never share passwords or OTPs.

Cyber Security Measures:

- The bank uses email filtering to detect phishing emails.
- Anti-phishing systems block malicious links.
- Multi-factor authentication (MFA) prevents unauthorized account access even if the password is stolen.

- **Example 2: Ransomware Attack on an Organization**
- A hospital's computer systems are infected with ransomware that encrypts patient records and demands payment to restore access.
- **Digital Safety Practice:**
- Employees avoid opening unknown email attachments.
- Staff receive training to recognize suspicious emails.
- **Cyber Security Measures:**
- Firewalls and antivirus software detect malicious files.
- Regular backups allow recovery without paying the ransom.
- Incident response teams isolate infected systems.

Example 3: Fake QR Code Scam

Fraudsters place fake QR code stickers over genuine payment QR codes in shops. Customers scan the fake code and send money to the scammer.

Digital Safety Practice:

Verify the merchant's name before confirming payment.

Check whether the QR code appears tampered with.

Confirm payment details before sending money.

Cyber Security Measures:

Payment apps use encryption and fraud detection.

Banks monitor unusual transactions and may block suspicious payments

Example 4: Social Media Account Protection

A user receives a message saying, "You've won a free smartphone. Click here to claim."

Digital Safety Practice:

- Ignore suspicious offers.
- Do not share login credentials.
- Enable two-factor authentication.
- Review privacy settings regularly.

Cyber Security Measures:

- Social media platforms detect fake accounts.
- AI systems identify spam messages.
- Login alerts notify users of suspicious access attempts.

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- **Cyber Security Measures:**
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